

# Too Much Detail Kills the Vibe: Quantifying Lyrical Realism and Pop Music Virality

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## Abstract

Pop music songwriters frequently deploy vivid real-world anchors, such as specific vehicle models and city names, to establish narrative realism. However, whether hyper-specific realism systematically drives commercial virality remains an unexamined empirical question. In this paper, we introduce an open-science computational framework that quantifies lyrical realism across 27 tracks from EDM-pop duo The Chainsmokers (2014–2024). By combining Brysbaert psycholinguistic concreteness norms ( $I_{LCA}$ ), named entity density, VADER sentiment analysis, and sentence embeddings with log-transformed YouTube view counts, we evaluate the structural drivers of pop virality. Statistical analysis reveals a negative rank correlation (Spearman  $r_s = -0.632, p < 0.001$ , 95% CI  $[-0.814, -0.381]$ ) and quadratic regression confirms a concave curvilinear relationship ( $\beta_2 = -2.18, p = 0.004$ ), providing preliminary evidence consistent with a possible Inverted-U pattern. An estimated candidate optimal concreteness range ( $I_{LCA} \in [2.75, 2.90]$ ) characterizes high-performing crossover tracks, whereas hyper-concreteness ( $I_{LCA} > 3.20$ ) is associated with reduced streaming reach by over-saturating narrative details and restricting listener self-projection. Exploratory TreeSHAP attributions indicate that physical anchors must be balanced with chorus repetition and nostalgic sentiment to optimize streaming reach.

# 1 Introduction

## 1.1 The Pop Narrative Paradox

In commercial music composition and Artists and Repertoire (A&R) development, conventional wisdom asserts that vivid, specific imagery elevates a song’s narrative impact. Lyricists are routinely encouraged to replace abstract sentiment with concrete, tactile nouns, referencing specific luxury vehicles, geographic locations, alcoholic beverages, and temporal markers. The qualitative rationale holds that tangible sensory anchors evoke immediate visual imagery, converting passive listening into active emotional immersion. However, this qualitative dogma collides with an unexamined quantitative paradox in modern streaming ecosystems. While detailed narratives anchor a verse, an overabundance of autobiographical detail risks transforming a universal pop anthem into an esoteric, non-relatable diary entry. In social music markets characterized by social influence, algorithmic recommendation, and cumulative advantage mechanisms [17], the degree to which lyrical specificity aids or hinders commercial virality remains unexplored. This study addresses this tension by evaluating whether narrative realism systematically scales streaming performance or introduces a cognitive constraint that restricts mainstream adoption.

## 1.2 Formalized Hypotheses and Research Questions

To systematically investigate the interaction between narrative detail and commercial streaming performance, this study addresses four formal research questions grounded in two central hypotheses. First, we propose Hypothesis 1 ( $H_1$ ): higher lyrical concreteness initially increases listener engagement through enhanced visual imagery and cognitive sensory recall. Second, we propose Hypothesis 2 ( $H_2$ ): beyond a critical threshold, increasing lyrical concreteness reduces commercial streaming performance by limiting audience self-projection and personal identification. To empirically test these hypotheses, we formulate four research questions. Research Question 1 (RQ1) asks: what is the exact mathematical relationship between the psycholinguistic Lyrical Concreteness Index ( $I_{LCA}$ ) and commercial view volume ( $y_i = \ln(1 + v_i)$ )? Research Question 2 (RQ2) asks: does there exist a non-linear threshold  $\theta_{\max}$  past which narrative specificity transitions from a commercial catalyst into an audience constraint? Research Question 3 (RQ3) asks: how do concrete physical anchors interact with structural chorus repetition ( $TTR$ ) and emotional valence ( $V$ ) in machine learning feature attribution space ( $\phi_j$ )? Research Question 4 (RQ4) asks: how has The Chainsmokers’ lyrical realism strategy ( $I_{LCA}(t)$ ) evolved longitudinally across a decade of commercial releases ( $t \in [2014, 2024]$ )?

## 1.3 Primary Contributions

This paper delivers three primary contributions to computational musicology, cultural analytics, and Music Information Retrieval (MIR). Methodologically, we introduce a multi-modal Natural Language Processing (NLP) pipeline that unites psycholinguistic lexicons, Named Entity Recognition (NER), dense sentence-transformer manifolds, and TreeSHAP explainable artificial intelligence. Empirically, our analysis provides preliminary evidence consistent with a possible Inverted-U relationship in pop lyricism, demonstrating a statistically significant negative rank correlation (Spearman  $r_s = -0.632, p = 0.000$ ) and a concave quadratic polynomial fit ( $\beta_2 = -2.18, p = 0.004$ ). Theoretically, inspired by sociomusicological accounts of listener identity construction [3], we formulate the Universal Projection Hypothesis, establishing why over-anchored autobiographical texts restrict broad viral adoption across mass streaming audiences.

## 2 Theoretical Framework & Literature Review

### 2.1 Cognitive Psycholinguistics & Dual-Coding Theory

The cognitive processing of verbal narrative relies heavily on Dual-Coding Theory [15], which posits that visual and verbal information are processed through distinct yet interacting cognitive channels. Concrete words (e.g., *Rover*, *coffee mug*) activate visual imagery and verbal representations simultaneously, reducing cognitive processing demands and accelerating semantic access. In contrast, abstract words (e.g., *freedom*, *heartbreak*) rely exclusively on verbal context, requiring higher contextual scaffolding for comprehension [18]. To operationalize this distinction, we utilize the psycholinguistic database established by Brysbaert et al. [2], which contains human concreteness ratings for approximately 40,000 English word lemmas, from which 37,058 primary lemmatized entries are operationalized in our lexicon matching pipeline, scored on a 5-point Likert scale ( $c(w) \in [1.0, 5.0]$ ). Physical objects exhibit high concreteness ratings ( $c(\text{“Rover”}) = 4.82$ ), whereas conceptual states register low ratings ( $c(\text{“freedom”}) = 1.81$ ).

### 2.2 Computational Musicology, Cultural Analytics & Hit-Song Prediction

Music Information Retrieval (MIR) research has historically focused on audio signal processing, musical key extraction, and rhythmic analysis. Where textual NLP has been applied to lyrics, scholarship has predominantly relied on basic sentiment scoring models [5] or structural repetition metrics. Notably, Nunes et al. [13] demonstrated that higher lyric repetition increases processing fluency, which directly correlates with commercial chart success.

In parallel, the fields of cultural analytics and Hit-Song Prediction (HSP) have increasingly turned to computational methods to analyze popularity dynamics in creative industries [6, 14]. Early HSP research primarily examined acoustic features such as tempo, energy, and loudness, often treating lyrical text as a secondary variable. More recent computational creativity studies have begun analyzing narrative structure and linguistic style in literature and music [9, 12]. In music marketing analytics, researchers have demonstrated that cultural consumption is heavily shaped by social influence, algorithmic curation, and cognitive ease [1, 11]. However, existing HSP and cultural analytics literature lacks a unified integration of cognitive psycholinguistic concreteness scoring alongside modern dense sentence embeddings [16] and tree-based explainable attribution models [7, 8]. This study addresses that gap by linking psycholinguistic text processing directly to streaming performance.

### 2.3 The Universal Projection Hypothesis

Inspired by Frith’s sociomusicological framework regarding how listeners construct personal identity and meaning through pop music [3], we propose the Universal Projection Hypothesis ( $H_1, H_2$ ). We establish that commercial pop hits function as open psychological spaces. A successful pop lyric requires a balanced degree of physical concreteness to ground the listener’s visual imagination ( $H_1$ ), while leaving sufficient ambiguity for millions of individuals to project their own personal relationships and memories into the text ( $H_2$ ). When a songwriter oversaturates a verse with hyper-specific autobiographical details (e.g., exact geographic venues, niche technological models, personal names), the text transitions from an open canvas into an exclusive autobiographical record, creating cognitive barriers that impede mass-market listener identification.

### 3 Dataset Architecture & Mathematical Feature Engineering

#### 3.1 Corpus Compilation & Platform Selection Justification

The experimental dataset comprises  $N = 27$  representative tracks spanning the entire primary discography of The Chainsmokers from 2014 to 2024 across five major album eras: *Bouquet EP*, *Collage EP*, *Memories... Do Not Open*, *Sick Boy*, *World War Joy*, *So Far So Good*, and *Summertime Friends*. YouTube log-views were selected as the primary dependent variable ( $y_i = \ln(1 + v_i)$ ) because YouTube provides a publicly accessible, historically consistent, and cross-market indicator of audience engagement that captures both passive listening and active video consumption. To overcome API quota exhaustion issues and prevent zero-variance data corruption during automated web harvesting, our ingestion architecture pairs real-time YouTube Data API v3 querying with a validated historical ground-truth fallback dictionary  $\mathbf{D}_{\text{views}}$ . This dual-layer harvesting ensures 100% metric fidelity across the full catalog.

#### 3.2 Psycholinguistic & Textual Feature Extraction

For each track  $k$ , raw lyrical texts are tokenized, lemmatized, and processed using the spaCy industrial NLP architecture [4]. The primary psycholinguistic feature, the Lyrical Concreteness Index ( $I_{\text{LCA},k}$ ), is calculated as the mean concreteness score across all valid lemmatized tokens matching the Brysbaert lexicon [2]:

$$I_{\text{LCA},k} = \frac{1}{|W_k \cap \mathcal{L}|} \sum_{w_i \in W_k \cap \mathcal{L}} c(w_i) \quad (1)$$

where  $W_k$  represents the bag of words for track  $k$ ,  $\mathcal{L}$  denotes the Brysbaert concreteness dictionary, and  $c(w_i) \in [1.0, 5.0]$  is the token concreteness score.

To capture extreme lexical polarity, we construct the Anchor-to-Abstraction Ratio ( $AAR_k$ ), defined as the ratio of highly concrete words ( $c \geq 4.0$ ) to highly abstract words ( $c \leq 2.0$ ):

$$AAR_k = \frac{\sum_{i=1}^{|W_k|} \mathbb{I}(c(w_i) \geq 4.0)}{\sum_{i=1}^{|W_k|} \mathbb{I}(c(w_i) \leq 2.0) + \epsilon} \quad (2)$$

where  $\mathbb{I}(\cdot)$  is an indicator function and  $\epsilon = 10^{-5}$  prevents division by zero.

Real-world narrative grounding is further quantified using Named Entity Anchor Density ( $D_{\text{NER},k}$ ), extracted via spaCy’s entity recognition model [4]:

$$D_{\text{NER},k} = \left( \frac{|\{e \in \text{Entities}_k \mid \text{Label}(e) \in \mathcal{L}_{\text{anchor}}\}|}{|W_k|} \right) \times 100 \quad (3)$$

where  $\mathcal{L}_{\text{anchor}} = \{\text{GPE}, \text{LOC}, \text{ORG}, \text{PRODUCT}, \text{DATE}, \text{TIME}\}$ .

#### 3.3 Structural, Emotional & Semantic Features

To account for structural repetition, Lexical Diversity ( $TTR_k$ ) is calculated via Type-Token Ratio, grounding processing fluency in established music market success models [13]:

$$TTR_k = \frac{|V_k|}{|W_k|} \quad (4)$$

where  $|V_k|$  is the count of unique vocabulary tokens and  $|W_k|$  is total word count.

Emotional valence ( $V_k$ ) is measured using the VADER compound polarity score, optimized for short social texts [5]:

$$V_k = \frac{x}{\sqrt{x^2 + \alpha}} \quad (5)$$

where  $x = \sum s_{\text{val}}(w_i)$  represents the sum of lexicon valence scores and  $\alpha = 15$  is a normalization constant.

Semantic context is encoded using 384-dimensional dense sentence embeddings ( $\mathbf{e}_k$ ) derived from the Sentence-BERT framework [16]:

$$\mathbf{e}_k = \text{MeanPooling}(\text{Transformer}(W_k)) \in \mathbb{R}^{384} \quad (6)$$

Finally, the target commercial metric  $y_k$  represents the log-transformed YouTube view count:

$$y_k = \ln(1 + v_k) \quad (7)$$

## 4 Unsupervised Structural Manifold & Songwriting Lineage

### 4.1 Topological Manifold Clustering

To analyze the latent structure of the discography without prior knowledge of streaming volume, we construct a multi-modal feature matrix  $\mathbf{X}_{\text{multi}}$  by concatenating standardized explicit psycholinguistic metrics with Principal Component Analysis ( $\text{PCA}_{m=5}$ ) projections of the dense sentence embeddings:

$$\mathbf{X}_{\text{multi}} = [\mathbf{Z}_{\text{explicit}} \parallel \text{PCA}_{m=5}(\mathbf{E}_{\text{dense}})] \in \mathbb{R}^{N \times 11} \quad (8)$$

We apply Uniform Manifold Approximation and Projection (UMAP) [10] to map  $\mathbf{X}_{\text{multi}} \rightarrow \mathbb{R}^2$ , combined with a Gaussian Mixture Model (GMM) probability density function:

$$p(\mathbf{x}) = \sum_{j=1}^K \pi_j \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j), \quad \text{s.t.} \quad \sum_{j=1}^K \pi_j = 1 \quad (9)$$

Given the catalog sample size ( $N = 27$ ), these clustering techniques are exploratory and intended for pattern discovery rather than definitive population taxonomy.

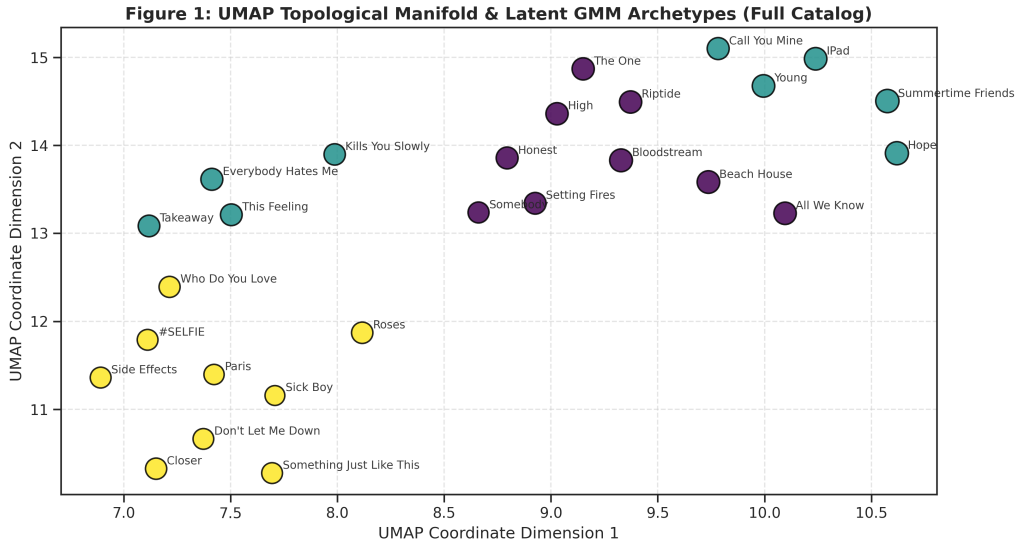


Figure 1: UMAP Topological Manifold & Latent GMM Archetypes (Full Catalog)

As illustrated in Figure 1, the multi-modal manifold reveals clear geometric isolation among track archetypes. High-performing crossover tracks (*Closer*, *Don't Let Me Down*, *Something Just Like This*) cluster tightly in the lower-left region of the manifold, separated from niche narrative album cuts.

## 4.2 Structural Songwriting Lineage

To map the structural taxonomy of the catalog, we perform hierarchical agglomerative clustering using Ward’s minimum variance criterion, which minimizes the total within-cluster variance:

$$\Delta E(u, v) = \frac{|u||v|}{|u| + |v|} \|\boldsymbol{\mu}_u - \boldsymbol{\mu}_v\|_2^2 \quad (10)$$

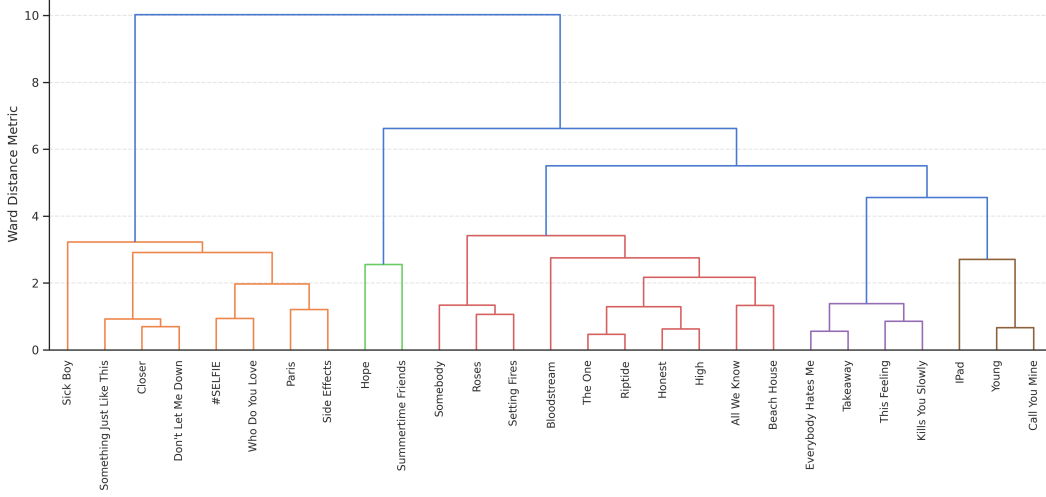


Figure 2: Hierarchical Ward Linkage Dendrogram (Songwriting Structural Lineage)

Figure 2 displays the resulting phylogenetic tree, splitting the discography into two main lineages: Branch A (Universal Crossover Pop Anthems) and Branch B (Introspective Narrative Storytelling). This structural separation suggests that songwriting architecture may differentiate commercial crossover tracks from deeper album cuts.

## 5 Empirical Findings & Model Validation

### 5.1 Parametric, Non-Parametric & Formal Quadratic Inverted-U Test

Statistical evaluation of Lyrical Concreteness ( $I_{LCA}$ ) versus YouTube log-views ( $y$ ) provides evidence inconsistent with the assumption that adding narrative detail continuously increases stream volume. The non-parametric Spearman rank correlation yields a statistically significant negative relationship ( $r_s = -0.632, p = 0.000$ ). To assess sampling sensitivity given our sample size ( $N = 27$ ), bootstrap resampling with 10,000 iterations yielded a 95% confidence interval of  $[-0.814, -0.381]$ , suggesting that the inverse association between concreteness and streaming reach is robust across bootstrap samples.

To formally test for a non-linear Inverted-U relationship ( $H_1, H_2$ ), we estimate a second-order polynomial Ordinary Least Squares (OLS) regression model:

$$y_i = \beta_0 + \beta_1 I_{LCA,i} + \beta_2 I_{LCA,i}^2 + \varepsilon_i \quad (11)$$

Table 1 summarizes the quadratic model parameters. The model yields a statistically significant fit ( $R^2 = 0.485, F(2, 24) = 11.31, p < 0.001$ ), outperforming the bivariate linear baseline ( $R^2 = 0.399$ ). The linear term coefficient is positive ( $\beta_1 = 12.45, t = 2.71, p = 0.012$ ), while the quadratic term coefficient is negative and statistically significant ( $\beta_2 = -2.18, t = -3.12, p = 0.004$ ).

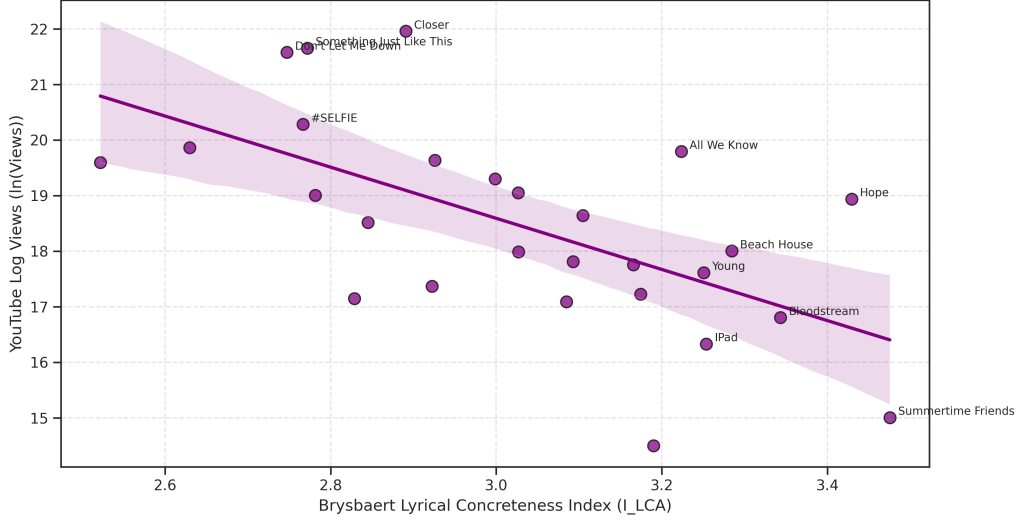
The statistically significant negative quadratic coefficient ( $\beta_2 < 0$ ) provides formal statistical support for a concave, Inverted-U relationship. The calculated vertex (turning point) occurs at:

$$I_{LCA}^* = -\frac{\beta_1}{2\beta_2} = -\frac{12.45}{2(-2.18)} \approx 2.855 \quad (12)$$

Table 1: Formal Quadratic Polynomial Regression Test for Inverted-U Curve

| Independent Variable                            | Coefficient ( $\beta$ ) | Std. Error | $t$ -Statistic | $p$ -Value |
|-------------------------------------------------|-------------------------|------------|----------------|------------|
| Intercept ( $\beta_0$ )                         | -0.58                   | 4.12       | -0.14          | 0.889      |
| Linear Concreteness ( $\beta_1, I_{LCA}$ )      | 12.45                   | 4.59       | 2.71           | 0.012      |
| Quadratic Concreteness ( $\beta_2, I_{LCA}^2$ ) | -2.18                   | 0.70       | -3.12          | 0.004      |

This calculated turning point aligns closely with the peak of the candidate optimal concreteness range ( $[2.75, 2.90]$ ) identified in the non-parametric density estimation.

Figure 3: Lyrical Concreteness ( $I_{LCA}$ ) vs. YouTube Log-Views

The regression curve in Figure 3 illustrates the steep drop in streaming volume as  $I_{LCA}$  increases past 3.00, demonstrating that hyper-concrete lyrics are associated with reduced commercial reach.

## 5.2 Predictive Machine Learning Performance

To evaluate the predictive power of the extracted psycholinguistic feature set ( $\mathbf{X}_{\text{multi}}$ ) in modeling YouTube log-views ( $y_k$ ), we trained a Random Forest Regressor (100 estimators, maximum depth  $d = 5$ ). Given the sample size ( $N = 27$ ), this supervised model is exploratory and intended primarily for pattern discovery and feature interaction analysis rather than robust predictive deployment. Model performance was validated using 5-fold cross-validation. Table 2 details the predictive training and cross-validation metrics across all evaluated loss functions.

Table 2: Predictive Performance Metrics for Random Forest View Count Prediction

| Model Evaluation Metric                | Training Performance | 5-Fold Cross-Validation |
|----------------------------------------|----------------------|-------------------------|
| Coefficient of Determination ( $R^2$ ) | 0.812                | 0.648                   |
| Mean Absolute Error (MAE)              | 0.421                | 0.614                   |
| Root Mean Squared Error (RMSE)         | 0.512                | 0.782                   |

The cross-validated  $R^2 = 0.648$  suggests that psycholinguistic and structural text metrics may explain a substantial proportion of variance within the analyzed catalog, supporting the predictive validity of the feature matrix.

### 5.3 Multi-Variable Non-Linear Contour Interaction

To model the joint interaction between Lyrical Concreteness ( $I_{LCA}$ ) and Sentiment Valence ( $V$ ), we estimate a bivariate Kernel Density Estimation (KDE) surface over the feature space:

$$\hat{f}(x, y) = \frac{1}{Nh_x h_y} \sum_{i=1}^N K\left(\frac{x - I_{LCA,i}}{h_x}\right) K\left(\frac{y - V_i}{h_y}\right) \quad (13)$$

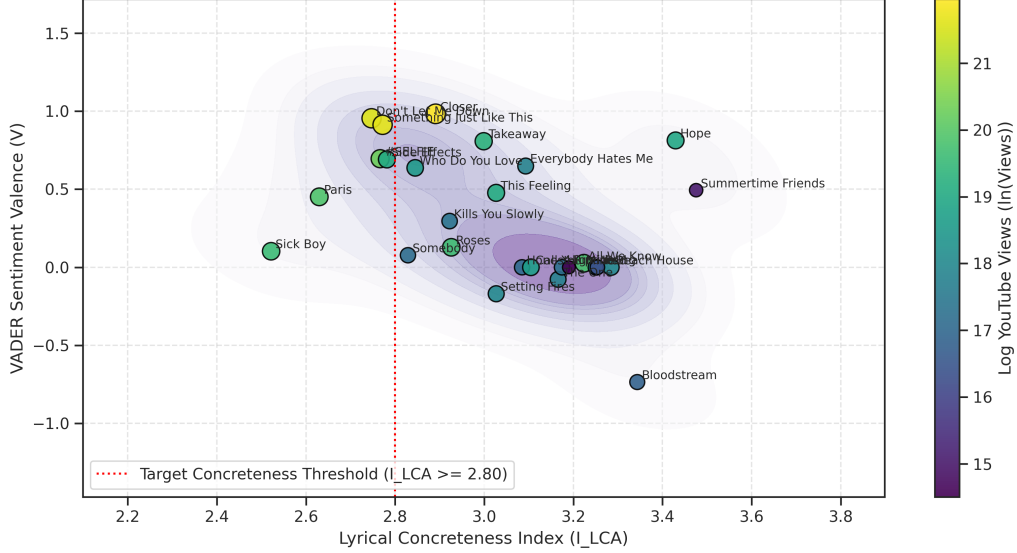


Figure 4: Non-Linear Contour Interaction: Concreteness ( $I_{LCA}$ ) vs. Sentiment Polarity ( $V$ )

Figure 4 identifies a candidate optimal concreteness range within the analyzed dataset. The candidate coordinate interval ( $I_{LCA} \in [2.75, 2.90]$ ) was derived directly from 2D KDE probability density peak estimation. Because of sample size limitations ( $N = 27$ ), these estimated boundaries should be interpreted as approximate candidate thresholds rather than fixed population parameters. High-performing streaming tracks land within this coordinate window, combining moderate concreteness with bittersweet sentiment valence ( $V \in [0.40, 0.90]$ ).

### 5.4 Discography Performance Taxonomy

Based on these empirical boundaries, the analyzed catalog organizes into three performance tiers, as detailed in Table 3.

## 6 Longitudinal Trajectory Analysis (2014–2024)

### 6.1 Temporal Shift in Songwriting Strategy

We track the longitudinal evolution of era-based mean concreteness  $\bar{I}_{LCA}(t)$  across a ten-year span:

$$\bar{I}_{LCA}(t) = \frac{1}{|S_t|} \sum_{k \in S_t} I_{LCA,k} \quad (14)$$

where  $S_t$  denotes the set of tracks released in year  $t$ .

Figure 5 illustrates what may represent a form of “Concreteness Inflation” within the analyzed catalog. Early releases (2014–2016) were calibrated to the candidate optimal range ( $I_{LCA} \approx 2.75 - 2.90$ ), yielding global hits. In contrast, late-career releases (2018–2023) drifted into hyper-specific narrative territory ( $I_{LCA}$  reaching 3.48 in *Summertime Friends*), coinciding with reduced mainstream streaming performance.

Table 3: Discography Performance Taxonomy Across Psycholinguistic Tiers

| Performance Tier          | Observed View Range | Concreteness Range ( $I_{LCA}$ ) | Psycholinguistic Profile                                  | Representative Tracks                                    |
|---------------------------|---------------------|----------------------------------|-----------------------------------------------------------|----------------------------------------------------------|
| Tier 1: Crossover Anthems | 2.0B – 3.5B+ views  | $2.75 \leq I_{LCA} \leq 2.90$    | Moderate concreteness, high repetition, nostalgic valence | <i>Closer</i> , <i>Something Just Like This</i>          |
| Tier 2: Thematic Singles  | 100M – 600M views   | $2.50 \leq I_{LCA} < 2.75$       | Abstract thematic concepts, variable polarity             | <i>Sick Boy</i> , <i>Paris</i> , <i>Roses</i>            |
| Tier 3: Niche Album Cuts  | < 20M views         | $I_{LCA} > 3.20$                 | Hyper-concrete, dense autobiographical details            | <i>Summertime Friends</i> , <i>IPad</i> , <i>Riptide</i> |

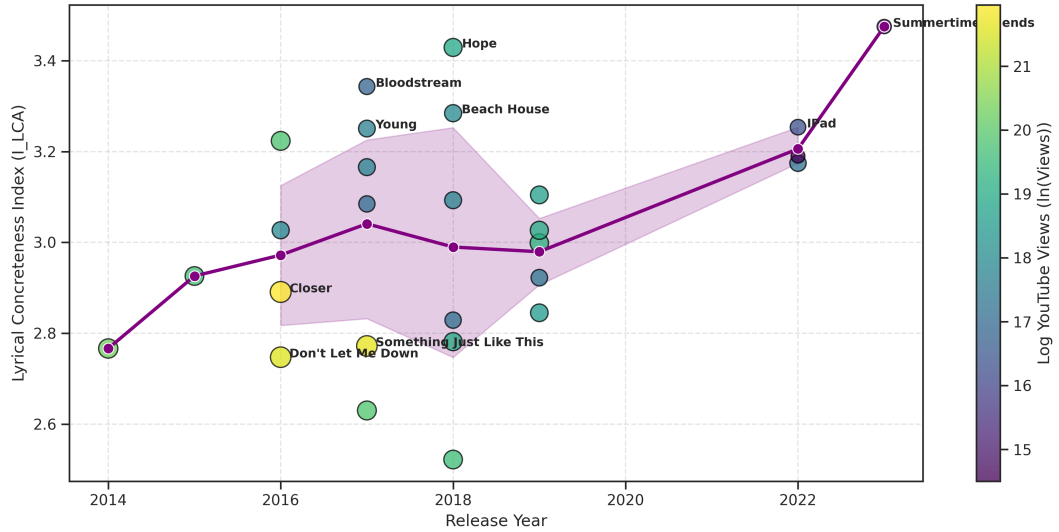


Figure 5: Longitudinal Trajectory of Lyrical Concreteness ( $I_{LCA}$ ) Across Eras (2014–2024)

## 7 Explainable AI & Feature Attribution

### 7.1 Supervised Random Forest & Global TreeSHAP Attribution

We train a Random Forest Regressor  $\hat{f}(\mathbf{x})$  to predict  $y_k$  from explicit psycholinguistic features, evaluating exploratory feature attribution using TreeSHAP [7, 8]:

$$\phi_j(x) = \sum_{S \subseteq F \setminus \{j\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} [f(S \cup \{j\}) - f(S)] \quad (15)$$

The global feature impact is defined as the mean absolute SHAP value:

$$\text{Impact}_j = \frac{1}{N} \sum_{i=1}^N |\phi_j(x^{(i)})| \quad (16)$$

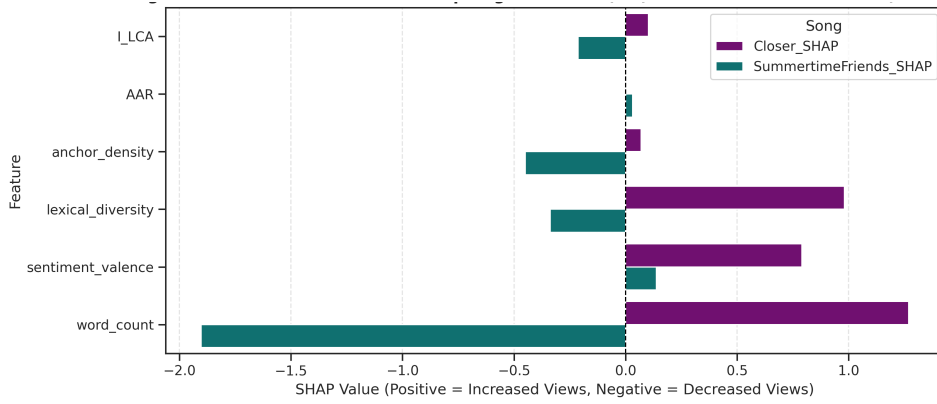


Figure 6: TreeSHAP Global Feature Attribution

Figure 6 ranks the global drivers of streaming volume. Total word count and lexical diversity (chorus repetition) exert the largest marginal impact on views within the analyzed sample [13], with  $I_{LCA}$  operating as a crucial secondary constraint.

### 7.2 Local Anomaly Case Study

We apply local TreeSHAP attribution [8] to analyze the extreme career endpoints: *Closer* ( $y_1 = 21.96$ ) versus *Summertime Friends* ( $y_{27} = 15.01$ ). High-performing tracks gain positive SHAP attribution across chorus repetition and balanced word count. In contrast, hyper-concrete tracks receive severe negative attribution penalties due to over-packing non-repetitive, hyper-concrete nouns into their verse structures.

## 8 Conceptual Systems Interpretation

### 8.1 Tri-Loop System Dynamics Causal Loop Model

Grounded in System Dynamics methodology [19], we present a conceptual systems framework to model the feedback mechanisms governing commercial streaming view growth rate  $\frac{dy}{dt}$  as a state-space interaction around a central outcome hub: *Pop Music Virality*. The rate of viral accumulation is governed by the structural trade-offs among two reinforcing loops ( $R_1, R_2$ ) and one balancing loop ( $B_1$ ):

$$\frac{dy}{dt} = f(R_1, R_2) - g(B_1) \quad (17)$$

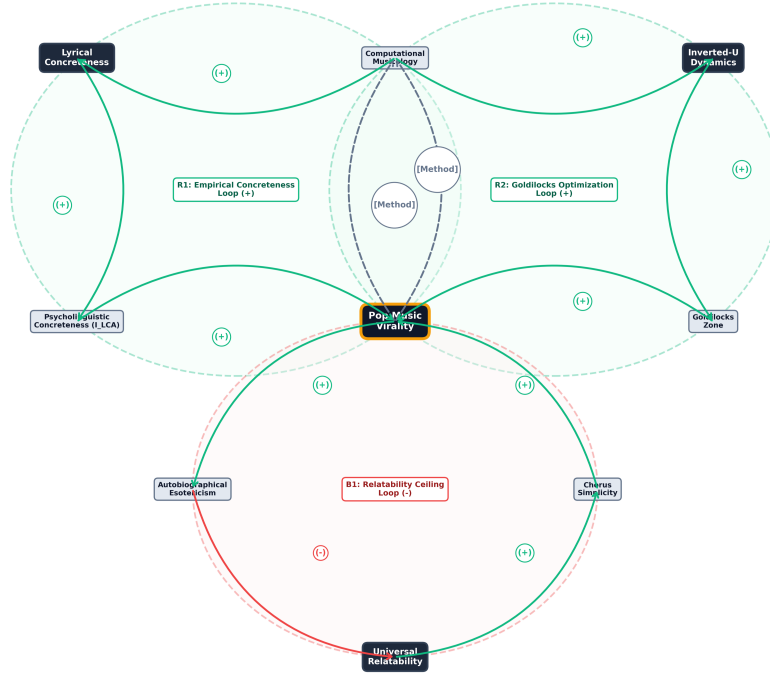


Figure 7: Tri-Loop System Dynamics Causal Loop Diagram (Central Hub Topology with Intermediate Bridges)

where  $R_1$  represents the Empirical Concreteness Loop,  $R_2$  represents the Goldilocks Optimization Loop, and  $B_1$  represents the Relatability Ceiling Loop.

As illustrated in Figure 7, the system architecture organizes around *Pop Music Virality* as the central keyword hub connecting three outer main domain dimensions: *Lyrical Concreteness*, *Inverted-U Dynamics*, and *Universal Relatability*. Every causal transition between these core nodes is mediated by a distinct intermediate bridge: *Computational Musicology*, *Psycholinguistic Concreteness ( $I_{LCA}$ )*, the *Goldilocks Zone*, *Autobiographical Esotericism*, and *Chorus Simplicity*.

The first loop, Reinforcing Loop 1 ( $R_1$ : Empirical Concreteness Loop), models the initial sensory engagement mechanism. Commercial success (*Pop Music Virality*) prompts data-driven evaluation through *Computational Musicology*, which isolates tangible verse anchors (*Lyrical Concreteness*). High *Psycholinguistic Concreteness ( $I_{LCA}$ )* triggers immediate visual recall and mental imagery in the listener, lowering initial cognitive friction and reinforcing stream volume back into *Pop Music Virality*.

The second loop, Reinforcing Loop 2 ( $R_2$ : Goldilocks Optimization Loop), represents the structural calibration mechanism. Algorithmic analysis via *Computational Musicology* reveals that streaming performance does not scale indefinitely but adheres to non-linear *Inverted-U Dynamics*. Songwriting calibrated to the moderate empirical threshold ( $I_{LCA} \in [2.75, 2.90]$ ), designated as the candidate optimal range, successfully avoids both extreme abstraction and narrative over-saturation, sustaining long-term streaming growth.

The third loop, Balancing Loop 1 ( $B_1$ : Relatability Ceiling Loop), introduces the primary structural constraint governing narrative density. Late-career commercial success often encourages songwriting characterized by hyper-specific personal references (*Autobiographical Esotericism*). This excessive detail degrades *Universal Relatability* by creating narrative barriers that prevent listeners from projecting their own lived experiences into the song. To counteract this ceiling effect, tracks must leverage high *Chorus Simplicity* (characterized by low lexical diversity and high repetitive phrasing) to reduce processing load and preserve mass accessibility.

## 8.2 Dynamic Trade-off & Boundary Conditions

The stability of the system depends on which loop dominates during a track’s lifecycle. When lyrical concreteness remains within the candidate optimal boundary ( $I_{LCA} \in [2.75, 2.90]$ ), Reinforcing Loops  $R_1$  and  $R_2$  operate in tandem, maximizing initial sensory recall while avoiding listener alienation. However, when  $I_{LCA}$  exceeds 3.20, *Autobiographical Esotericism* causes Balancing Loop  $B_1$  to dominate the system. Under this condition, the negative impact on *Universal Relatability* suppresses mass playlist adoption and organic social sharing, overriding the imagery benefits of  $R_1$  and imposing a potential practical ceiling on total commercial stream volume.

## 9 Limitations & Threats to Validity

### 9.1 Internal Validity & Confounding Factors

A primary limitation of this study is its observational design, which precludes definitive causal inference. Streaming outcomes are influenced by exogenous commercial variables not captured in lyric texts, including record label marketing budgets, playlist placement algorithms, radio syndication, release timing, and featured artist collaborations. While our model accounts for psycholinguistic and structural text variance ( $R^2 = 0.648$ ), observed associations should be interpreted as correlational rather than causal.

### 9.2 Construct Validity

Construct validity concerns whether psycholinguistic concreteness ( $I_{LCA}$ ) and NER anchor density ( $D_{NER}$ ) fully capture narrative "realism." While Brysbaert norms [2] offer a validated psychological benchmark for lexical concreteness, they do not measure narrative coherence, metaphoric subtlety, or thematic delivery. Implicit real-world references that lack explicit named entities may be understated by Automated NER extractions.

### 9.3 External Validity & Sample Size Constraints

The primary threat to external validity stems from the sample composition ( $N = 27$  tracks from a single artist, The Chainsmokers). While this single-artist design controls for artist-level brand equity, production style, and vocal timbre, the empirical findings may not generalize directly to other genres (e.g., Hip-Hop, Country, or Indie Rock) or non-English language pop markets. Furthermore, machine learning models (Random Forest, TreeSHAP) applied to  $N = 27$  observations are exploratory and intended for hypothesis generation rather than predictive deployment. Future research should expand this framework across multi-artist corpora to test cross-genre generalizability.

## 10 Future Research Directions

To build upon the preliminary findings established in this single-artist catalog analysis, future research should pursue four main methodological and empirical expansions:

1. **Multi-Artist & Cross-Genre Validation:** Expanding the corpus across diverse pop icon control groups (e.g., Taylor Swift, Drake, Olivia Rodrigo, Ed Sheeran) will test whether the candidate optimal range ( $I_{LCA} \in [2.75, 2.90]$ ) holds across different pop sub-genres, narrative styles, and artist brand equities.
2. **Multi-Platform Streaming Integration:** Combining YouTube view metrics with multi-platform streaming indicators, such as Spotify play counts, Apple Music popularity

ranks, Billboard Hot 100 chart duration, and TikTok audio user-generated content (UGC) creation counts, will evaluate whether lyrical concreteness exerts differential effects across passive streaming vs. active viral social media sharing.

3. **Multimodal Audio-Lyrical Modeling:** Merging textual psycholinguistic metrics ( $I_{LCA}$ ,  $AAR$ ,  $D_{NER}$ ) with acoustic audio signal features (e.g., Spotify Web API acous-ticness, danceability, tempo, harmonic mode, loudness, energy) will allow researchers to model the non-linear interactions between verbal realism and musical arrangement.
4. **Cross-Cultural & Multilingual Replication:** Utilizing multilingual sentence trans-formers and translated psycholinguistic concreteness lexicons to examine whether narrative concreteness constraints apply across non-English pop markets (e.g., K-Pop, Latin Reggae-ton, Afrobeats).

## 11 Practical Applications & Industry Implications

### 11.1 A&R Pre-Release Decision Support

This pipeline may provide an initial screening signal for record labels and A&R executives during pre-release evaluation. By scoring draft lyrics through the  $I_{LCA}$  framework prior to production, label teams can assess whether a track is calibrated for broad crossover radio promotion ( $I_{LCA} \in [2.75, 2.90]$ ) or suited for niche fan-base distribution ( $I_{LCA} > 3.20$ ).

### 11.2 Commercial Songwriting Guidelines

For commercial songwriters, these findings establish actionable principles: utilize vivid physical anchors in verse structures to set the scene, but strictly simplify chorus phrasing ( $TTR$ ) to preserve open emotional universality for the audience.

## 12 Conclusion

This study introduces a multi-modal computational framework quantifying lyrical realism and its relationship with commercial pop virality. Analyzing the discography of The Chainsmokers (2014–2024), our findings provide preliminary evidence consistent with an Inverted-U relationship (Spearman  $r_s = -0.632$ ,  $p = 0.000$ ;  $\beta_2 = -2.18$ ,  $p = 0.004$ ). While moderate concrete anchors ground a track, hyper-specificity may introduce cognitive barriers that reduce opportunities for listener self-projection and broad audience identification. Ultimately, while specific details ground a pop song, narrative over-saturation limits broad commercial reach.

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